SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	AARTHI SREE.G	**Reg. No.:**	192572047
Section / Batch:	CSE(AI)	**Date:**	10/07/2026

Code of Conduct

I AARTHI SREE.G/192572047____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: AARTHI SREE.G______

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
		GRAND TOTAL:	**100 Marks**

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G 192572047RA

QUESTIONS

- Q1 C Code – Array Traversal 10 marks
- Q2 C Code – Linear Search 10 marks
- Q3 C Code – Binary Search Mid 10 marks
- Q4 C Code – Pointer with Array 10 marks
- Q5 C Code – Array Update 10 marks
- Q6 C Code - Linked List Node 10 marks
- Q7 C Code - Time Complexity Loop 10 marks
- Q8 C Code - Nested Loop 10 marks
- Q9 C Code - Binary Search Complexity 10 marks
- Q10 C Code - Pointer Increment 10 marks

Q9 **C Code - Binary Search Complexity** 10 marks

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

0(log n)

Save Answer

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G 192572047RA

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

- Q1 C Code – Array Traversal 10 marks
- Q2 C Code – Linear Search 10 marks
- Q3 C Code – Binary Search Mid 10 marks
- Q4 C Code – Pointer with Array 10 marks
- Q5

Q3 **C Code – Binary Search Mid** 10 marks

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Your Answer

3

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G 192572047RA

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

- Q1 C Code – Array Traversal 10 marks
- Q2 C Code – Linear Search 10 marks
- Q3 C Code – Binary Search Mid 10 marks
- Q4 C Code – Pointer with Array 10 marks
- Q5 C Code – Array Update 10 marks
- Q6 C Code - Linked List Node 10 marks
- Q7 C Code - Time Complexity Loop 10 marks

Q1 **C Code – Array Traversal** 10 marks

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Your Answer

50

Save Answer

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G
192572047RA

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7
C Code - Time Complexity Loop
10 marks

Q4 **C Code – Pointer with Array** 10 marks

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Your Answer

30

Save Answer

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G
192572047RA

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7
C Code - Time Complexity Loop
10 marks

Q5 **C Code – Array Update** 10 marks

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

Your Answer

4

Save Answer

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G
192572047RA

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7
C Code - Time Complexity Loop
10 marks

Q6 **C Code - Linked List Node** 10 marks

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G
192572047RA

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1 C Code – Array Traversal 10 marks
Q2 C Code – Linear Search 10 marks
Q3 C Code – Binary Search Mid 10 marks
Q4 C Code – Pointer with Array 10 marks
Q5 C Code – Array Update 10 marks
Q6 C Code - Linked List Node 10 marks
Q7 C Code - Time Complexity Loop 10 marks

Q7 **C Code - Time Complexity Loop** 10 marks

```
for(i=0;i<n;i++)
printf("*");
The complexity is
```

Your Answer

0(n)

Save Answer

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1 C Code – Array Traversal 10 marks
Q2 C Code – Linear Search 10 marks
Q3 C Code – Binary Search Mid 10 marks
Q4 C Code – Pointer with Array 10 marks
Q5 C Code – Array Update 10 marks
Q6 C Code - Linked List Node 10 marks
Q7 C Code - Time Complexity Loop 10 marks

Q8 **C Code - Nested Loop** 10 marks

```
for(i=0;i<n;i++)
for(j=0;j<n;j++)
printf("*");
```

Your Answer

0(n)

Save Answer

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

AARTHI SREE G
192572047RA

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1 C Code – Array Traversal 10 marks
Q2 C Code – Linear Search 10 marks
Q3 C Code – Binary Search Mid 10 marks
Q4 C Code – Pointer with Array 10 marks
Q5 C Code – Array Update 10 marks
Q6 C Code - Linked List Node 10 marks
Q7 C Code - Time Complexity Loop 10 marks

Q9 **C Code - Binary Search Complexity** 10 marks

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

0(log n)

Save Answer

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7
C Code - Time Complexity Loop
10 marks

Q10 **C Code - Pointer Increment**

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10

Save Answer